

江苏省专转本 2023 届高等数学真题试卷

答案解析

1. 选 A

解: $\because x=1$ 是函数的第一类间断点

$$\therefore (x-a)|_{x=1} = 0, \text{ 即 } a=1, \lim_{x \rightarrow 1} \frac{x^2+x+b}{x-1} \text{ 存在}$$

$$\text{因为 } \lim_{x \rightarrow 1} (x-1) = 0, \text{ 所以必有 } \lim_{x \rightarrow 1} (x^2+x+b) = 2+b=0, \text{ 所以 } b=-2$$

$$\text{故 } ab = -2$$

2. 选 C

$$\text{解: } f(x) - \lim_{x \rightarrow 0} f(x) = \frac{e^x - 1}{x} - \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \frac{e^x - 1}{x} - 1$$

$$\text{因为当 } x \rightarrow 0 \text{ 时, } f(x) - \lim_{x \rightarrow 0} f(x) = \frac{e^x - 1}{x} - 1 \text{ 与 } ax^k \text{ 为等价无穷小}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\frac{e^x - 1}{x} - 1}{ax^k} = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{ax^{k+1}} = \lim_{x \rightarrow 0} \frac{e^x - 1}{a(k+1)x^k} = \lim_{x \rightarrow 0} \frac{1}{a(k+1)x^{k-1}} = 1$$

$$\text{所以 } k-1=0, \quad k=1, \quad a(k+1)=1, \quad a=\frac{1}{2}$$

3. 选 B

$$\text{解: } y' = \frac{1}{1+x}, \quad y'' = \frac{-1}{(1+x)^2}, \quad y''' = \frac{1 \cdot 2}{(1+x)^3}, \quad y^{(4)} = \frac{-1 \cdot 2 \cdot 3}{(1+x)^4}$$

$$\therefore f^{(2023)}(x) = \frac{2022!}{(1+x)^{2023}}, \text{ 故 } f^{(2023)}(0) = 2022!$$

4. 选 D

解: $\because f(x)$ 的一个原函数为 $\sin 2x$

$$\therefore f(x) = (\sin 2x)' = 2 \cos 2x, \quad f'(x) = -4 \sin 2x, \quad f'(1) = -4 \sin 2$$

5. 选 C

解: $f(x+y, x-y) = x^2 - y^2$, 即 $f(x+y, x-y) = (x+y)(x-y)$

$$f(u, v) = uv$$

$$f'_u(u, v) = v, \therefore f'_u(2, 0) = 0$$

$$f'_v(u, v) = u, \therefore f'_v(2, 0) = 2$$

6. 选 B

解: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^p}$ 当 $0 < p \leq 1$ 时条件收敛

$\sum_{n=1}^{\infty} \frac{1}{p^n}$ 当 $0 < p \leq 1$ 时 $\lim_{n \rightarrow \infty} \frac{1}{p^n} \neq 0$ 发散

7. 选 A

$$\text{解: } A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 1 & a & 1 \\ a & 1 & 1 \\ 1 & 1 & a \end{pmatrix} \xrightarrow{\substack{r_2 - ar_1 \\ r_3 - r_1}} \begin{pmatrix} 1 & a & 1 \\ 0 & 1-a^2 & 1-a \\ 0 & 1-a & a-1 \end{pmatrix}$$

$$\therefore R(A) = 2, \text{ 所以应有 } \frac{1-a^2}{1-a} = \frac{1-a}{a-1}, \text{ 解得 } a = -2$$

8. 选 D

$$\text{解: } (\alpha_1 + k\alpha_2, \alpha_2 + k\alpha_3, \alpha_3 + k\alpha_1) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 & k \\ k & 1 & 0 \\ 0 & k & 1 \end{pmatrix}$$

$$\text{要使 } \alpha_1 + k\alpha_2, \alpha_2 + k\alpha_3, \alpha_3 + k\alpha_1 \text{ 线性相关, 则应有 } \begin{vmatrix} 1 & 0 & k \\ k & 1 & 0 \\ 0 & k & 1 \end{vmatrix} = 0$$

$$\text{解得 } k = -1$$

9. 解: 因为极限值是一个常数, 所以设 $\lim_{x \rightarrow 0} f(x) = A$

于是 $f(x) = x \sin \frac{1}{x} + \frac{1}{x} \sin 2x - A$, 两边取 $x \rightarrow 0$ 的极限

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} + \frac{1}{x} \sin 2x \right) - A = 2 - A$$

$$2A = 2, \quad A = 1, \quad \text{所以 } \lim_{x \rightarrow 0} f(x) = 1$$

10. 解: $\because y = 3x + m$ 是曲线的切线, $\therefore k_{\text{切}} = 3$

$$\frac{dx}{dt} = 2t - 1, \quad \frac{dy}{dt} = 2t + 1$$

$$\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{2t+1}{2t-1} = 3, \quad \text{解得 } t = 1$$

$t = 1$ 对应点处坐标为 $(1, 1)$, 所以切线方程为 $y - 1 = 3(x - 1)$

即 $y = 3x - 2$, 所以 $m = -2$

11. 解: $\because f(x)$ 是连续的奇函数, $\therefore f(0) = 0$

$$\text{于是 } \lim_{n \rightarrow \infty} n f\left(\frac{2}{n}\right) = \lim_{n \rightarrow \infty} \frac{f\left(\frac{2}{n}\right) - f(0)}{\frac{2}{n}} \cdot 2 = f'(0) \cdot 2 = 1$$

$$\therefore f'(0) = \frac{1}{2}$$

12. 解: $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^3}}, \quad \frac{dx}{dy} = \sqrt{1+x^3}, \quad \left. \frac{dx}{dy} \right|_{x=1} = \sqrt{2}$

13. 解: 令 $\lim_{n \rightarrow \infty} \left| \frac{(x+1)^{n+1}}{(n+1)a^{n+1}} \cdot \frac{na^n}{(x+1)^n} \right| = \frac{|x+1|}{|a|} < 1$

$$\text{所以 } |x+1| < |a|, \quad -1 - |a| < x < -1 + |a|, \quad a = \pm 2$$

当 $a = -2$ 时, 在 $x = -3$ 处级数为 $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 舍去

当 $a = 2$ 时, 在 $x = -3$ 处级数为 $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ 收敛

所以 $a = 2$

14. 解: $|AB| = |A||B| = 4|B| = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{vmatrix} = 20$, $\therefore |B| = 5$

15. 解: $\lim_{x \rightarrow 0} \frac{x \sin x + 2 \cos x - 2}{x^2 - x^2 \cos x} = \lim_{x \rightarrow 0} \frac{x \sin x + 2 \cos x - 2}{\frac{1}{2}x^4}$

$$= \lim_{x \rightarrow 0} \frac{\sin x + x \cos x - 2 \sin x}{2x^3}$$

$$= \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{2x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{6x^2} = -\frac{1}{6}$$

16. 解: $\int x f(x) dx = x^2 \arctan x + c$ 两边求导

$$x f(x) = 2x \arctan x + \frac{x^2}{1+x^2}, \therefore f(x) = 2 \arctan x + \frac{x}{1+x^2}$$

$$\begin{aligned} \int f(x) dx &= 2 \int \arctan x dx + \int \frac{x}{1+x^2} dx \\ &= 2x \arctan x - 2 \int \frac{x}{1+x^2} dx + \int \frac{x}{1+x^2} dx \\ &= 2x \arctan x - \frac{1}{2} \int \frac{1}{1+x^2} d(1+x^2) \\ &= 2x \arctan x - \frac{1}{2} \ln(1+x^2) + c \end{aligned}$$

17. 解: 令 $x = 2 \sin t$, 当 $x = 0$ 时 $t = 0$, 当 $x = \sqrt{2}$ 时 $t = \frac{\pi}{4}$

$$\text{原式} = \int_0^{\frac{\pi}{4}} \frac{4 \sin^2 t}{2 \cos t} d(2 \sin t) = 4 \int_0^{\frac{\pi}{4}} \sin^2 t dt = 2 \int_0^{\frac{\pi}{4}} (1 - \cos 2t) dt$$

$$= (2t - \sin 2t) \Big|_0^{\frac{\pi}{4}} = \frac{\pi}{2} - 1$$

18. 解: 方程两边对 x 求偏导数

$$3z^2 \cdot \frac{\partial z}{\partial x} - 3z - 3x \frac{\partial z}{\partial x} = 0 \quad \therefore z^2 \frac{\partial z}{\partial x} - z - x \frac{\partial z}{\partial x} = 0 \quad (1)$$

$$\frac{\partial z}{\partial x} = \frac{z}{z^2 - x}$$

方程两边对 y 求偏导数

$$3z^2 \cdot \frac{\partial z}{\partial y} - 3x \frac{\partial z}{\partial y} + 6 = 0, \therefore \frac{\partial z}{\partial y} = \frac{-2}{z^2 - x}$$

①式两边再对 y 求偏导数

$$2z \cdot \frac{\partial z}{\partial y} \cdot \frac{\partial z}{\partial x} + z^2 \frac{\partial^2 z}{\partial x \partial y} - \frac{\partial z}{\partial y} - x \frac{\partial^2 z}{\partial x \partial y} = 0$$

由原方程可知, 当 $x=0, y=0$ 时, $z=1$, $\left. \frac{\partial z}{\partial x} \right|_{\substack{x=0 \\ y=0}} = 1$, $\left. \frac{\partial z}{\partial y} \right|_{\substack{x=0 \\ y=0}} = -2$

$$\text{所以 } \left. \frac{\partial^2 z}{\partial x \partial y} \right|_{\substack{x=0 \\ y=0}} = 2$$

19. 解: $y = xe^x$ 是 $y'' + py' + qy = -e^x$ 的一个特解

$\therefore y = xe^x$ 代入微分方程中两边应相等

$$\text{即 } (x+2)e^x + p(x+1)e^x + qxe^x = -e^x$$

比较两边同类项可知, $p = -3, q = 2$

微分方程为 $y'' - 3y' + 2y = -e^x$

对应齐次方程的特征方程 $r^2 - 3r + 2 = 0$, 特征根 $r_1 = 1, r_2 = 2$

其通解 $\bar{y} = c_1 e^x + c_2 e^{2x}$

原方程通解 $y = c_1 e^x + c_2 e^{2x} + xe^x$

由 $y|_{x=0} = 1$ 得 $c_1 + c_2 = 1$ ①

$$y' = c_1 e^x + 2c_2 e^{2x} + (x+1)e^x$$

由 $y'|_{x=0} = 2$ 得 $c_1 + 2c_2 + 1 = 2$ ②

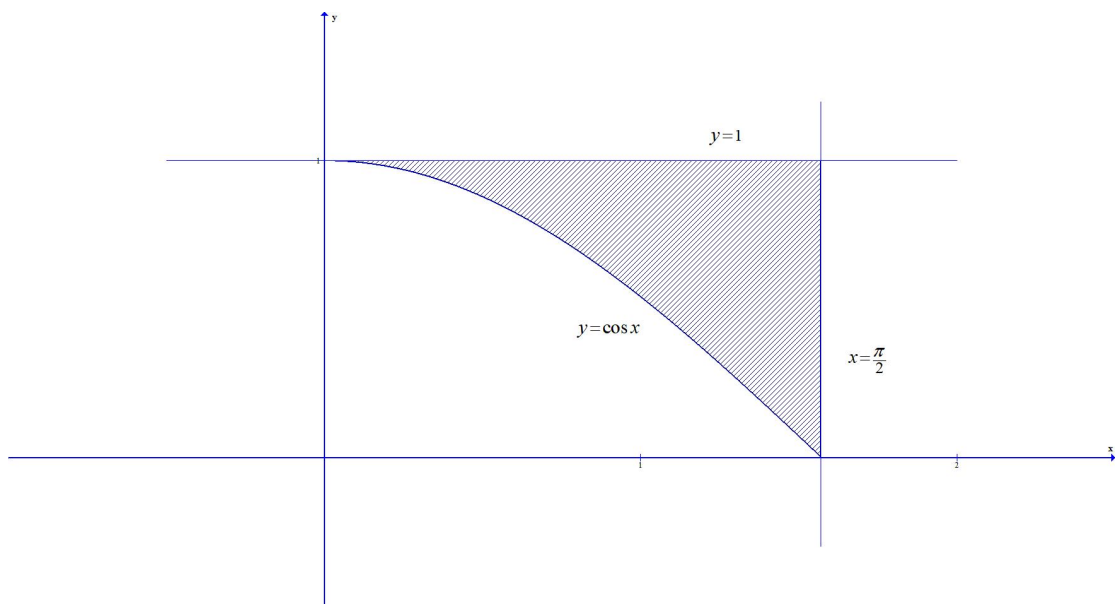
由①②解得 $c_1 = 1, c_2 = 0$

所求特解为 $y = e^x + xe^x$

$$20. \text{ 解: } \iint_D (2y - \cos x) dx dy = \int_0^{\frac{\pi}{2}} dx \int_{\cos x}^1 (2y - \cos x) dy$$

$$= \int_0^{\frac{\pi}{2}} dx (y^2 - y \cos x) \Big|_{\cos x}^1$$

$$= \int_0^{\frac{\pi}{2}} (1 - \cos x) dx = \frac{\pi}{2} - 1$$



21. 解: 由 $AX = A - X$ 得 $X = (A + E)^{-1} A$

$$(A + E \quad E) = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[r_3+r_1]{r_2-2r_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow[r_1-r_3]{r_1-r_2} \begin{pmatrix} 1 & 0 & 0 & 2 & -1 & -1 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

$$(A + E)^{-1} = \begin{pmatrix} 2 & -1 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$X = (A + E)^{-1} A = \begin{pmatrix} 2 & -1 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

22. 解: $\bar{A} = \begin{pmatrix} 1 & -2 & 0 & 1 \\ 2 & -3 & 1 & \lambda+1 \\ 1 & -1 & 1 & \lambda^2 \end{pmatrix} \xrightarrow[r_3-r_1]{r_2-2r_1} \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & \lambda-1 \\ 0 & 1 & 1 & \lambda^2-1 \end{pmatrix}$

$$\xrightarrow{r_3-r_2} \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & \lambda-1 \\ 0 & 0 & 0 & \lambda(\lambda-1) \end{pmatrix}, \text{ 当 } \lambda = 0 \text{ 或 } \lambda = 1 \text{ 时方程组有解}$$

当 $\lambda = 0$ 时, 方程组为
$$\begin{cases} x_1 - 2x_2 = 1 \\ 2x_1 - 3x_2 + x_3 = 1 \\ x_1 - x_2 + x_3 = 0 \end{cases}$$

$$\bar{A} = \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1+2r_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

x_1, x_2 为约束变量, x_3 为自由变量

一般解 $\begin{cases} x_1 = -1 - 2x_3 \\ x_2 = -1 - x_3 \end{cases}$, 令 $x_3 = 0$ 得原方程组的一个特解 $\eta^* = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

导出组一般解 $\begin{cases} x_1 = -2x_3 \\ x_2 = -x_3 \end{cases}$, 令 $x_3 = 1$ 得 $\xi = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

所以原方程通解为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} + k \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ (其中 k 为任意常数)

当 $\lambda = 1$ 时, 方程组为
$$\begin{cases} x_1 - 2x_2 = 1 \\ 2x_1 - 3x_2 + x_3 = 2 \\ x_1 - x_2 + x_3 = 1 \end{cases}$$

$$\bar{A} = \begin{pmatrix} 1 & -2 & 0 & 1 \\ 2 & -3 & 1 & 2 \\ 1 & -1 & 1 & 1 \end{pmatrix} \xrightarrow[r_3-r_1]{r_2-2r_1} \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{r_3-r_2} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

x_1, x_2 为约束变量, x_3 为自由变量

一般解 $\begin{cases} x_1 = 1 - 2x_3 \\ x_2 = -x_3 \end{cases}$, 令 $x_3 = 0$ 得原方程组的一个特解 $\eta^* = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

导出组一般解 $\begin{cases} x_1 = -2x_3 \\ x_2 = -x_3 \end{cases}$, 令 $x_3 = 1$ 得 $\xi = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

所以原方程通解为 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + k \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ (其中 k 为任意常数)

23. 证明: 令 $F(x) = \sin x + \cos x - 1 - x - x^2$, $F(0) = 0$

$$F'(x) = \cos x - \sin x - 1 + 2x, \quad F'(0) = 0$$

$$F''(x) = -\sin x - \cos x + 2$$

当 $x > 0$ 时, $F''(x) > 0$, 所以 $F'(x)$ 单调递增, $F'(x) > F'(0) = 0$

由 $F'(x) > 0$ 又可知 $F(x)$ 单调递增, $F(x) > F(0) = 0$

即 $\sin x + \cos x - 1 - x + x^2 > 0$, 故 $\sin x + \cos x > 1 + x - x^2$

24. 解: (1) $\because y = 0$ 是曲线的水平渐近线

$$\therefore \lim_{x \rightarrow \infty} \frac{ax^2 + 2x - 1}{x^2 + bx + 1} = 0, \quad \therefore a = 0$$

又 $\because x = 1$ 是曲线的垂直渐近线

$$\therefore (x^2 + bx + 1) \Big|_{x=1} = 0, \quad \therefore b = -2$$

$$(2) f(x) = \frac{2x-1}{(x-1)^2}, \quad \text{定义域 } (-\infty, 1), (1, +\infty)$$

$$f'(x) = \frac{2(x-1)^2 - (2x-1) \cdot 2(x-1)}{(x-1)^4} = \frac{-2x}{(x-1)^3}$$

$$f''(x) = \frac{-2(x-1)^3 + 2x \cdot 3(x-1)^2}{(x-1)^6} = \frac{4x+2}{(x-1)^4}$$

$$\text{令 } f''(x) = 0 \text{ 得 } x = -\frac{1}{2}$$

x	$\left(-\infty, -\frac{1}{2}\right)$	$-\frac{1}{2}$	$\left(-\frac{1}{2}, 1\right)$	$(1, +\infty)$
$f''(x)$	-	0	+	+
$f(x)$	$\cap$	拐点	$\cup$	$\cup$

由表可知, 函数 $f(x)$ 的凹区间为 $\left(-\frac{1}{2}, 1\right)$, $(1, +\infty)$, 凸区间为 $\left(-\infty, -\frac{1}{2}\right)$

$$\text{拐点} \left(-\frac{1}{2}, -\frac{8}{9}\right)$$

25. 解: $A(t) = \int_0^t f(x)dx$, 由 $A(t) = \frac{1}{2} - \frac{1}{2}f(t)$ 得

$$\int_0^t f(x)dx = \frac{1}{2} - \frac{1}{2}f(t), \text{ 两边求导得}$$

$$f(t) = -\frac{1}{2}f'(t), \text{ 即 } f'(t) + 2f(t) = 0$$

$$\text{通解 } f(t) = e^{-\int 2dt} \cdot c = ce^{-2t}$$

$\because$ 曲线过点 $(0,1)$, 所以 $c=1$

$$V(t) = \pi \int_0^{e^{-2t}} t^2 dy + \pi \int_{e^{-2t}}^1 \left(\frac{1}{2} \ln y \right)^2 dy = \pi t^2 y \Big|_0^{e^{-2t}} + \frac{\pi}{4} \int_{e^{-2t}}^1 \ln^2 y dy$$

$$= \pi t^2 e^{-2t} + \frac{\pi}{4} y \ln^2 y \Big|_{e^{-2t}}^1 - \frac{\pi}{4} \int_{e^{-2t}}^1 y d \ln^2 y$$

$$= -\frac{\pi}{2} \int_{e^{-2t}}^1 \ln y dy = -\frac{\pi}{2} y \ln y \Big|_{e^{-2t}}^1 + \frac{\pi}{2} \int_{e^{-2t}}^1 y d \ln y$$

$$= -\pi t e^{-2t} + \frac{\pi}{2} y \Big|_{e^{-2t}}^1 = \frac{\pi}{2} (-2t e^{-2t} + 1 - e^{-2t})$$

$$\lim_{t \rightarrow +\infty} V(t) = \lim_{t \rightarrow +\infty} \frac{\pi}{2} (-2t e^{-2t} + 1 - e^{-2t}) = \frac{\pi}{2}$$

